YIT – DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

PROJECT GUIDELINES

3rd Year AIML Students

Academic Project – Guide & Student Handbook

2026–27

1. Purpose of the Project

The project is intended to help students apply Artificial Intelligence and Machine Learning concepts to a practical problem. The emphasis is not only on building a software application, but on identifying a meaningful problem, designing a solution, applying an appropriate AI/ML method, evaluating the result, and documenting the complete engineering process.

Students are expected to demonstrate:

- Problem identification and research skills
- Requirement analysis and system design
- Appropriate use of AI/ML techniques
- Software development and integration
- Testing and quantitative evaluation
- Teamwork, project management and communication
- Technical documentation and individual understanding

2. Project Selection Principles

- The project must address a clearly defined, practical problem.
- The project should contain a meaningful AI/ML component; AI should not be added only for decoration.
- Students should explain what is novel, improved, automated or intelligently decided by their system.
- Projects should be feasible within the academic timeline and available resources.
- Avoid overused topics such as basic face-recognition attendance, generic chatbots, disease prediction and simple CRUD management systems unless there is a substantial and justified innovation.
- Use publicly available or legitimately collected datasets. Respect privacy, copyright and licensing requirements.

3. Recommended Project Architecture

Where applicable, projects should follow a flow similar to:

REAL-WORLD PROBLEM → DATA → PREPROCESSING → AI/ML → DECISION/MATCH/PREDICTION → APPLICATION → EVALUATION

4. Project Phases & Deliverables

Phase	Suggested Timeline	Main Deliverables
Phase 1 – Ideation & Proposal	Weeks 1–2	Proposal, Abstract, Problem Statement, Objectives, Scope
Phase 2 – Research	Weeks 3–4	Literature Survey, Existing vs Proposed System, Novelty
Phase 3 – Requirements	Weeks 5–6	SRS, Use Cases, Functional/Non-functional Requirements, RTM
Phase 4 – Design	Weeks 7–8	Architecture, UML, ER Diagram, Database Design, Data Dictionary
Phase 5 – AI/ML Planning	Weeks 9–10	Dataset Plan, Preprocessing, Model Selection, Evaluation Plan
Phase 6 – Development	Weeks 11–16	Working modules, integration, weekly progress log
Phase 7 – Testing	Weeks 17–18	Test Plan, Test Cases, AI/ML Evaluation, Defect Log
Phase 8 – Results & Finalization	Week 19	Results, Discussion, Final Report, User Manual

Phase 9 – Final Presentation	Week 20/21	PPT, Demo, Viva, Individual Contribution Report

5. Mandatory Project Documents

No.	Document	Minimum Contents
01	Project Proposal	Problem, motivation, objectives, scope, methodology, technologies
02	Abstract	200–300 word project summary
03	Literature Survey	8–15 relevant papers/projects with comparison
04	SRS	Functional and non-functional requirements, constraints, interfaces
05	System Design	Architecture, UML diagrams, data flow and module design
06	Database Design	ER diagram, tables and data dictionary, where applicable
07	AI/ML Design Document	Dataset, preprocessing, features, model, training and metrics
08	Test Plan & Test Cases	Unit, integration, system and AI/ML test cases
09	Final Project Report	Complete consolidated project documentation
10	PPT + Demo	Final presentation and working demonstration
11	Weekly Progress Log	Work completed, issues, decisions and next plan
12	Individual Contribution Report	Each member's responsibilities and contribution

6. SRS – Minimum Structure

1. Introduction: purpose, scope, users and definitions
2. Overall description
3. Functional requirements
4. Non-functional requirements
5. User roles and permissions
6. Hardware and software requirements
7. External interfaces
8. Constraints, assumptions and dependencies
9. Acceptance criteria

7. AI/ML Design – Minimum Structure

10. Problem formulation: classification, regression, recommendation, matching, detection, forecasting, etc.
11. Dataset source, size, classes/labels and collection method
12. Data cleaning and preprocessing
13. Feature extraction/representation
14. Model(s) considered and justification for final model
15. Training, validation and test strategy
16. Hyperparameters and important implementation choices
17. Evaluation metrics appropriate to the problem

18. Baseline/model comparison where feasible
19. Limitations, errors and possible improvements

8. Review Structure

Review	Students Must Demonstrate	Evidence
Review 1 – Proposal	Problem, motivation, literature, objectives, novelty and feasibility	Proposal + Literature Survey
Review 2 – Requirements & Design	SRS, architecture, UML, database, dataset and AI approach	SRS + Design Documents
Review 3 – Prototype	At least one complete end-to-end working flow	Working Prototype + Progress Log
Review 4 – AI & System Evaluation	Model results, testing, comparison, errors and improvements	Evaluation Report + Test Results
Final Review	Complete system, report, presentation, demo and individual viva	All Deliverables

9. Review Evaluation Criteria

Area	What the Guide Will Look For
Problem Understanding	Can clearly explain the problem, users and need
Literature & Novelty	Can identify existing work and explain the project's difference
Requirements & Design	Requirements and architecture are consistent and justified
AI/ML Understanding	Can explain data, model, metrics and limitations
Implementation	Working modules and integration
Testing	Systematic test cases and evidence
Results	Quantitative and meaningful results
Teamwork	Balanced and documented contribution
Presentation	Clear technical explanation and demonstration

10. Testing Requirements

At minimum, teams should include:

- Unit testing of important modules
- Integration testing between application and AI/database components
- System testing of major user workflows
- Negative/invalid input testing
- AI/ML evaluation using suitable metrics
- Evidence of failed cases and corrective action

11. Example Test Case Format

Test ID	Input/Condition	Expected Result	Actual Result	Status
TC-01	Valid login	Dashboard displayed	Dashboard displayed	Pass
TC-02	Invalid password	Error message	Error message	Pass
TC-03	Valid project input	AI result generated	AI result generated	Pass

12. Team Responsibilities

Each member must have a clearly defined technical responsibility. Example:

Role	Possible Responsibility
Member 1	AI/ML, dataset and model
Member 2	Backend, APIs and database
Member 3	Frontend/UI and integration
Member 4	Testing, evaluation, deployment/documentation

Roles may overlap, but every student must contribute technically. Every student must be able to explain the overall project and in detail explain their own contribution during viva.

13. Weekly Progress Log

Week	Work Completed	Problems/Decisions	Next Week Plan
1	Problem selection		
2	Literature survey		
3	Requirements		
4	Design		

Continue the same format for every project week. The guide may request the log at any review.

14. Final Report – Recommended Chapter Structure

20. Chapter 1 – Introduction
21. Chapter 2 – Literature Survey
22. Chapter 3 – Problem Statement, Objectives and Scope
23. Chapter 4 – Requirement Analysis
24. Chapter 5 – System Design
25. Chapter 6 – AI/ML Methodology
26. Chapter 7 – Implementation
27. Chapter 8 – Testing and Evaluation
28. Chapter 9 – Results and Discussion
29. Chapter 10 – Conclusion
30. Chapter 11 – Future Scope
31. References
32. Appendices

15. Academic & Technical Integrity

- Students must acknowledge datasets, papers, libraries, APIs and other external resources.
- Do not claim an AI result that was not actually obtained.
- Do not fabricate accuracy, dataset size, test results or screenshots.
- AI coding assistants may be used as learning/development aids where permitted, but students remain responsible for understanding and explaining the submitted code.
- Copied projects or substantially reproduced projects without acknowledgement may be rejected.

16. Final Submission Checklist

☐ Project Proposal

☐ Abstract

☐ Literature Survey

☐ SRS

☐ System Design + UML

☐ Database Design

☐ AI/ML Design Document

☐ Source Code

☐ Dataset / Dataset Reference

☐ Test Plan + Test Cases

☐ AI/ML Evaluation Results

☐ Final Report

☐ User Manual

☐ PPT

☐ Demo Video, if required

☐ Weekly Progress Log

☐ Individual Contribution Report

17. Suggested Evaluation Scheme

Component	Marks
Problem & Literature Study	10
SRS & Requirements	10
System/AI Design	15
Implementation & Integration	20
AI/ML Evaluation & Results	15
Testing	10
Documentation	10
Presentation, Demo & Viva	10

Total: 100 Marks. The final distribution may be modified to match departmental/university regulations.

18. Expectations from Students

- Meet the guide regularly and arrive prepared with measurable progress.
- Maintain version-controlled source code and regular backups.
- Do not wait until the final weeks to integrate the AI model with the application.
- Record important design decisions and changes in the progress log.
- Test early and repeatedly.
- Be able to demonstrate the project without relying entirely on one team member.

19. Final Message to Students

A good project is not defined by how complicated the AI model is. It is defined by how clearly you identify a real problem, design a sensible solution, apply AI appropriately, evaluate it honestly, and demonstrate that the system works. Focus on building something useful, explainable and technically sound.

— Department of Artificial Intelligence & Machine Learning —